Module 4.1: Probability Models, Expected Values, and Bayes' Formula

SCHWESERNOTES - BOOK 1

LOS 4.a: Calculate expected values, variances, and standard deviations and demonstrate their application to investment problems.

The **expected value** of a random variable is the weighted average of the possible outcomes for the variable. The mathematical representation for the expected value of random variable *X*, that can take on any of the values from x_1 to x_n, is:

$$E(X) = \Sigma P(x_i)x_i = P(x_1)x_1 + P(x_2)x_2 + \ldots + P(x_n)x_n$$

where:

$P(x_i)$ = probability of outcome x_i

Example: Expected earnings per share

The probability distribution of earnings per share (EPS) for Ron's Stores is given in the following figure. Calculate the expected EPS.

EPS Probability Distribution

Probability	EPS
10%	£1.80

20%	£1.60
40%	£1.20
30%	£1.00
100%	

Answer:

The expected EPS is simply a weighted average of each possible EPS, where the weights are the probabilities of each possible outcome:

$$E[EPS] = 0.10(1.80) + 0.20(1.60) + 0.40(1.20) + 0.30(1.00) = £1.28$$

Variance and *standard deviation* measure the dispersion of a random variable around its expected value, sometimes referred to as the **volatility** of a random variable. **Variance** (from a probability model) can be calculated as the probability-weighted sum of the squared deviations from the mean (or expected value). The standard deviation is the positive square root of the variance. The following example illustrates the calculations for a probability model of possible returns.

Example: Expected value, variance, and standard deviation from a probability model

Using the probabilities given in the following table, **calculate** the expected return on Stock A, the variance of returns on Stock A, and the standard deviation of returns on Stock A.

Event	Prob	R_A	Prob × R_A	R_A – E(R_A)	[R_A – E(R_A)]^2	Prob × [R_A – E(R_A)]^2
Boom	30%	20%	0.06	0.07	0.0049	0.00147
Normal	50%	12%	0.06	–0.01	0.0001	0.00005
Slow	20%	5%	0.01	–0.08	0.0064	0.00128
			E(R_A) = 0.13			Var (R_A) = 0.00280

Answer:

$$E(R_A) = (0.30 \times 0.20) + (0.50 \times 0.12) + (0.20 \times 0.05) = 0.13 = 13\%$$

The expected return for Stock A is the probability-weighted sum of the returns under the three different economic scenarios.

In Column 5, we have calculated the differences between the returns under each economic scenario and the expected return of 13%.

In Column 6, we squared all the differences from Column 5. In the final column, we have multiplied the probabilities of each economic scenario times the squared deviation of returns from the expected returns, and their sum, 0.00280, is the variance of R_A.

The standard deviation of $R_A = \sqrt{0.0028} = 0.0529.$

Note that in a previous reading, we estimated the standard deviation of a distribution from sample data, rather than from a probability model of returns. For the sample standard deviation, we divided the sum of the squared deviations from the mean by n – 1, where *n* was the size of the sample. Here, we have no "n" because we have no observations; a probability model is forward-looking. We use the probability weights instead, as they describe the entire distribution of outcomes.

LOS 4.b: Formulate an investment problem as a probability tree and explain the use of conditional expectations in investment application.

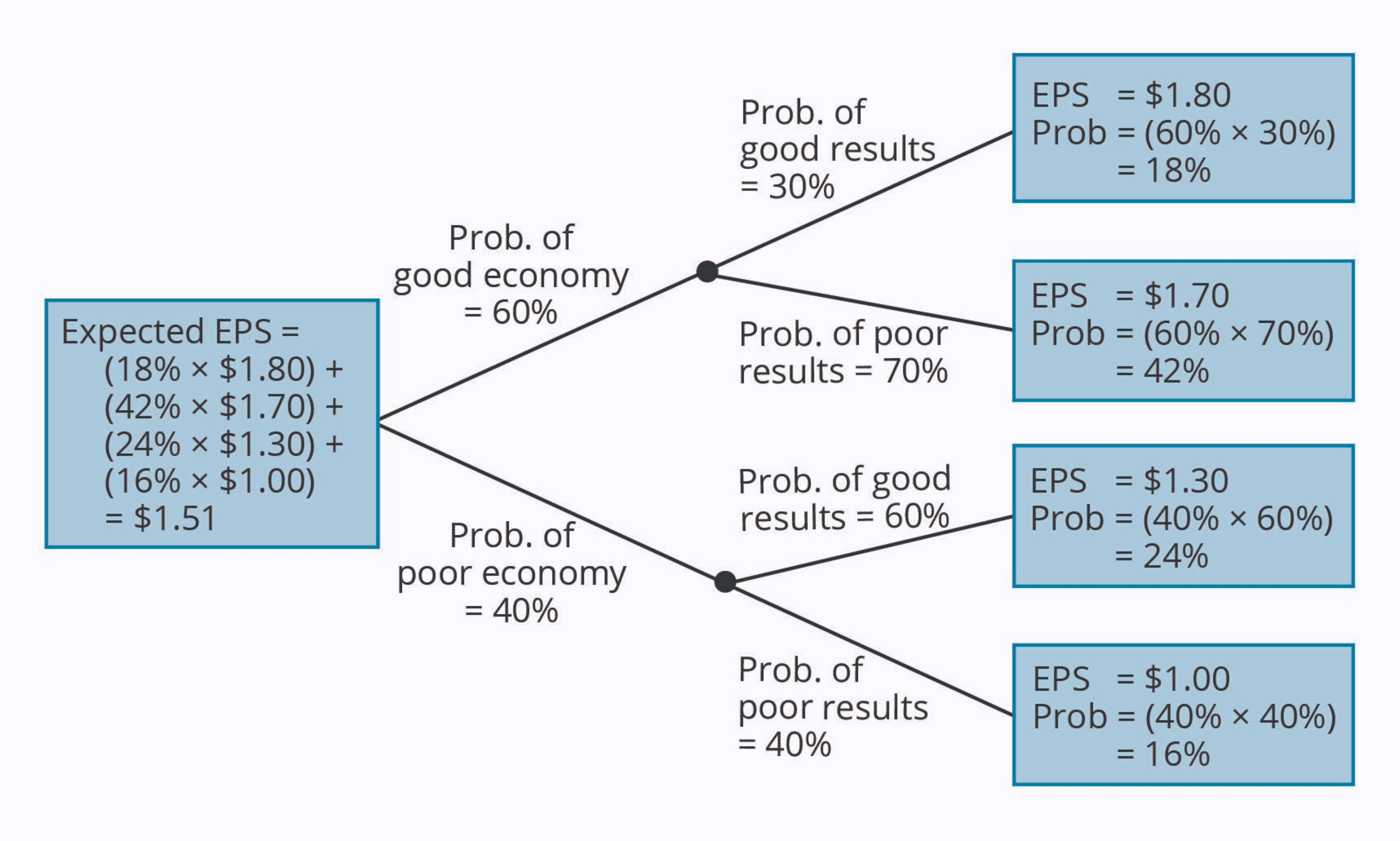
You may wonder where the returns and probabilities used in calculating expected values come from. A general framework, called a **probability tree**, is used to show the probabilities of various outcomes. In **A Probability Tree**, we have shown estimates of EPS for four different events: (1) a good economy and relatively good results at the company, (2) a good economy and relatively poor results at the company, (3) a poor economy and relatively good results at the company, and (4) a poor economy and relatively poor results at the company. Using the rules of probability, we can calculate the probabilities of each of the four EPS outcomes shown in the boxes on the right-hand side of the probability tree.

The expected EPS of $1.51 is simply calculated as follows:

$$(0.18 \times 1.80) + (0.42 \times 1.70) + (0.24 \times 1.30) + (0.16 \times 1.00) = \$1.51$$

Note that the probabilities of the four possible outcomes sum to 1.

A Probability Tree



Expected values or expected returns can be calculated using conditional probabilities. As the name implies, **conditional expected values** are contingent on the outcome of some other event. An analyst would use a conditional expected value to revise his expectations when new information arrives.

Consider the effect a tariff on steel imports might have on the returns of a domestic steel producer's stock in the previous example. The stock's conditional expected return, given that the government imposes the tariff, will be higher than the conditional expected return if the tariff is not imposed.

LOS 4.c: Calculate and interpret an updated probability in an investment setting using Bayes' formula.

Bayes' formula is used to update a given set of prior probabilities for a given event in response to the arrival of new information. The rule for updating prior probability of an event is as follows:

updated probability =

$$\frac{\text{probability of new information for a given event}}{\text{unconditional probability of new information}} \times \text{prior probability of event}$$

We can derive Bayes' formula using the multiplication rule and noting that P(AB) = P(BA):

P(B|A) × P(A) = P(BA), and P(A|B) × P(B) = P(AB)

Because P(BA) = P(AB), we can write P(B|A) P(A) = P(A|B) P(B), and $\frac{P(B|A)P(A)}{P(B)}$, equals $\frac{P(BA)}{P(B)}$, the joint probability of A and B divided by the unconditional probability of B.

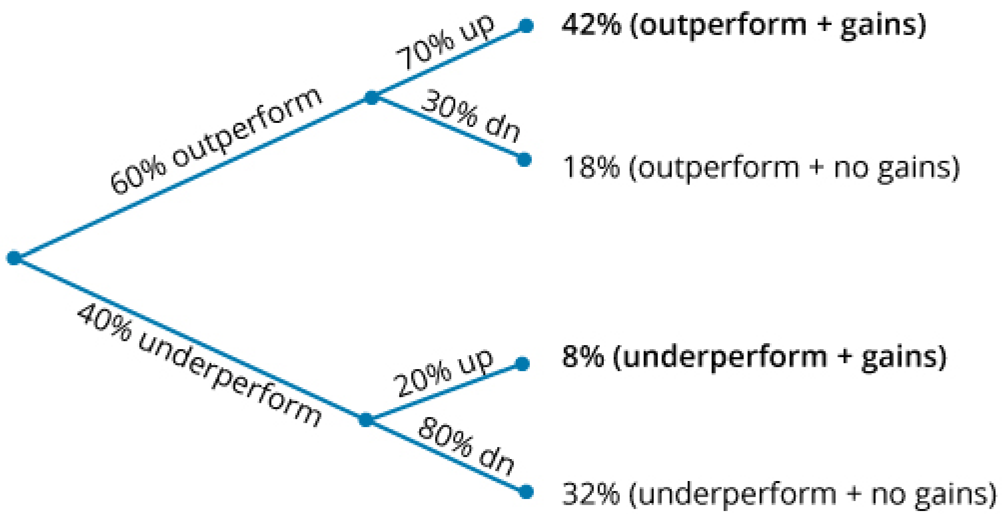
The following example illustrates the use of Bayes' formula. Note that A is *outperform* and A^C is *underperform*, P(BA) is (outperform + gains), P(A^CB) is (underperform + gains), and the unconditional probability P(B) is P(AB) + P(A^CB), by the total probability rule.

Example: Bayes' formula

There is a 60% probability that the economy will outperform—and if it does, there is a 70% probability a stock will go up and a 30% probability the stock will go down. There is a 40% probability the economy will underperform, and if it does, there is a 20% probability the stock in question will increase in value (have gains) and an 80% probability it will not. Given that the stock increased in value, calculate the probability that the economy outperformed.

Answer:

Bayes' Formula



In the **Bayes' Formula** just presented, we have multiplied the probabilities to calculate the probabilities of each of the four outcome pairs. Note that these sum to 1. Given that the stock has gains, what is our updated probability of an outperforming economy? We sum the probability of stock gains in both states (outperform and underperform) to get 42% + 8% = 50%. Given that the stock has gains and using Bayes' formula, the probability that the economy has outperformed is $\frac{42\%}{50\%} = 84\%$.

Reading 4: Key Concepts

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LOS 4.a

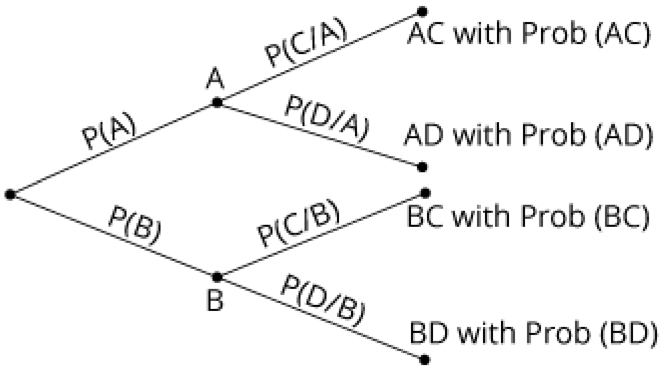
The expected value of a random variable is the weighted average of its possible outcomes:

$$E(X) = \Sigma P(x_i)x_i = P(x_1)x_1 + P(x_2)x_2 + \ldots + P(x_n)x_n$$

Variance can be calculated as the probability-weighted sum of the squared deviations from the mean or expected value. The standard deviation is the positive square root of the variance.

LOS 4.b

A probability tree shows the probabilities of two events and the conditional probabilities of two subsequent events:



Conditional expected values depend on the outcome of some other event. Forecasts of expected values for a stock's return, earnings, and dividends can be refined, using conditional expected values, when new information arrives that affects the expected outcome.

LOS 4.c

Bayes' formula for updating probabilities based on the occurrence of an event *O* is as follows:

$$P\left(I|O\right)=\frac{P(O|I)}{P(O)}\times P\left(I\right)$$

Equivalently, based on the following tree diagram, $P\left(A|C\right)=\dfrac{P(AC)}{P(AC)+P(BC)}$: